# Supplementary Information

Which spaceflight biology hardware can be modelled?

## Supplementary Note S1. Catalogue defects and source conflicts

SPLASH records #121–#147 carry descriptive fields displaced from neighbouring rows (offsets mostly −2). All fields of these records except the name, alternate names, background, category and status are excluded from every analysis. Table S1 lists the fields reviewed individually. Further defects found during curation: SPLASH #36 (MVP) lists "MOBIAS", a different bioreactor, as an alternate name; the OSDR registry links a ground cabinet of the Lunar Receiving Laboratory (registry 2052) to 36 unrelated studies through a catch-all payload, and links MOBIAS to the STS-115 Group Activation Pack experiment, which used hand-cranked packs; 310 of 641 OSDR datasets have no experiment-platform entry. Source conflicts recorded during literature curation are listed, per hardware item, in `hardware/CURATION_NOTES.md` and in the `note` field of each affected parameter.

Table S1: SPLASH fields judged misattributed, uncertain or duplicated (likely source row given).

| # | Field | Verdict | Source row | Reason |
|---|---|---|---|---|
| 127 | specimen_types_supported | misattributed | 125 | Succinonitrile is the transparent model alloy studied by PFMI (#125), not a protein-crystallization specimen |
| 128 | specimen_types_supported | misattributed | 126 | PCAM crystallization trays are the PCG-STES (#126) specimen format; BioMole is a molecular-biology (DNA/RNA) instrument |
| 128 | specimen_size | misattributed | 126 | 'up to 6 cylinders per STES' explicitly names STES (#126) |
| 129 | note_specs | misattributed | 127 | 'One Ryutai Rack holds ~6 cartridges' describes PCRF (#127), which is installed in the Ryutai Rack |
| 131 | specimen_size | uncertain | 129 | Syringe-pinned drop of ~25.4 mm matches Ring Sheared Drop (#129) rather than SUBSA ampoules |
| 132 | specimen_types_supported | misattributed | 130 | Solid combustible materials (PMMA, cotton) are SoFIE (#130) specimens, not solution crystals |
| 133 | specimen_types_supported | misattributed | 131 | Semiconductor/crystal melts are SUBSA (#131) specimens; SALI is a biological incubator |
| 133 | specimen_size | misattributed | 131 | Ampoules with a hot zone describe the SUBSA furnace (#131) |
| 134 | specimen_types_supported | uncertain | 132 | Generic solution samples; consistent with the -2 row offset seen in neighbouring records |
| 136 | specimen_types_supported | misattributed | 134 | 'Small live animals' cannot be a thermal-cycler (SmartCycler) specimen |
| 137 | specimen_types_supported | misattributed | 135 | Blood/saliva/urine/swab list belongs to the Wet Lab Kit (#135); identity of #137 is an aquatic habitat |
| 137 | specimen_size | misattributed | 135 | Vials, syringes, needles, gloves are Wet Lab Kit (#135) contents |
| 137 | background | duplicate_of | 69 | Background describes the JAXA Aquatic Habitat (AQH), which is also record #69 |
| 138 | specimen_types_supported | misattributed | 136 | RNA from microbial cultures describes the Wet Lab RNA SmartCycler (#136) |
| 138 | specimen_size | misattributed | 136 | 16 reaction tubes per run is a SmartCycler (#136) specification |
| 138 | background | duplicate_of | 34 | Record is named Plant Growth Unit but its background describes the Advanced Plant Habitat (#34) |

*Table S1, continued*

| # | Field | Verdict | Source row | Reason |
|---|---|---|---|---|
| 139 | specimen_types_ supported | misattributed | 137 | Medaka/zebrafish are Aquatic Habitat (#137/#69) specimens; AAU is a rodent transfer unit |
| 139 | specimen_size | misattributed | 137 | Two aquaria of ~0.7 L each describe the Aquatic Habitat |
| 140 | specimen_types_ supported | misattributed | 138 | Arabidopsis, dwarf wheat, lettuce, peppers are the APH crop list (#138 background = APH); BRIC-Opti is a small passive canister |
| 140 | specimen_size | misattributed | 138 | Shoot area ~1708 cm2 and 43 cm shoot height match APH; impossible in a BRIC canister |
| 141 | specimen_types_ supported | misattributed | 142 | Live adult mice belong to the Rodent Habitat (#142); ESA Biopack hosted cells and small organisms |
| 141 | specimen_size | misattributed | 139 | Mice in Transporter/Habitat and AAU transfers describe the rodent system (#139/#142) |
| 142 | specimen_types_ supported | misattributed | 140 | Microbes and plant tissue cultures in OptiCell plates describe a culture canister (BRIC-Opti #140), not the Rodent Habitat |
| 142 | specimen_size | misattributed | 140 | Up to 4 OptiCell plates is a BRIC-Opti (#140) capacity |
| 142 | note_specs | misattributed | 140 | Text names BRIC-Opti explicitly |
| 144 | specimen_types_ supported | misattributed | 142 | Mice belong to the rodent habitat; the Cell Culture Model hosts cells and microbes |
| 144 | specimen_size | misattributed | 142 | Up to 10 adult mice belongs to the rodent habitat |
| 145 | specimen_types_ supported | misattributed | 143 | Bacteria, fungi, cell cultures, small plants match BioRack (#143); AAEU housed aquatic animals |
| 146 | specimen_types_ supported | misattributed | 144 | Mammalian cell lines and microbial cultures match the Cell Culture Model (#144); FEU housed frogs |
| 146 | specimen_size | misattributed | 144 | CEU/CEU2/MEU chamber specifications name the Cell Culture Model (#144) units |
| 146 | temperature_range | misattributed | 144 | CEU/CEU2/MEU set points name the Cell Culture Model (#144) units |
| 147 | specimen_types_ supported | misattributed | 145 | Medaka, newt, toadfish were AAEU (#145) specimens; CEBAS flew swordtail fish, snails and hornwort |
| 123 | note_specs | misattributed | 121 | Text names MOBIAS (#121) explicitly |
| 120 | note_specs | uncertain | 123 | 5000x magnification at 10 kV describes an electron microscope, not a rack |
| 39 | specimen_size | uncertain | 42 | 'Type IV and Passive EUE' is EMCS (#42) experiment-container terminology |
| 39 | lighting | uncertain | 42 | 'EUE specific' is EMCS (#42) terminology |
| 89 | volume | uncertain | 91 | ELF is a JEM facility; 'EXPRESS rack drawer' unlikely, but non-biological |

# Supplementary Note S2. The MI-CFD-HW checklist

Table S2: The MI-CFD-HW checklist and its coverage across specimen-housing hardware (public sources).

| Item | Description | Group | Critical | Quant. (%) | Qual. (%) | Absent (%) |
|---|---|---|---|---|---|---|
| G1 | Enclosure dimensions | geometry | yes | 18 | 38 | 44 |
| G2 | Specimen-chamber volume or dimensions | geometry | yes | 17 | 52 | 31 |
| G3 | Specimen layout / number of units | geometry | | 28 | 9 | 63 |
| F1 | Ventilation mode (forced / passive / sealed) | flow | yes | 41 | 18 | 41 |
| F2 | Airflow rate or velocity | flow | yes | 4 | 0 | 96 |
| E1 | Temperature at specimen | environment | yes | 30 | 23 | 47 |
| E2 | Relative humidity | environment | | 13 | 14 | 74 |
| E3 | Gas composition / CO2-O2 control | environment | | 8 | 33 | 59 |
| E4 | Pressure | environment | | 11 | 4 | 86 |
| E5 | Lighting (type, flux or darkness) | environment | | 12 | 37 | 51 |
| B1 | Specimen types | biology | yes | 98 | 0 | 2 |
| B2 | Nutrient / water delivery | biology | | 34 | 28 | 38 |
| B3 | Specimen gas-exchange / metabolic load | biology | | 0 | 10 | 90 |
| T1 | Heat load / power | thermal | | 2 | 5 | 94 |
| W1 | Wall / membrane gas permeability | boundary | | 18 | 20 | 62 |

# Supplementary Note S3. Literature curation

Table S3: Critical MI-CFD-HW items after literature curation (✓sourced with a verbatim quotation; F2 n/a: sealed or passive hardware). G1 enclosure dimensions; G2 specimen container; F1 ventilation mode; F2 airflow; E1 temperature; B1 specimen.

| Hardware | G1 | G2 | F1 | F2 | E1 | B1 | Tier | sourced/params |
|---|---|---|---|---|---|---|---|---|
| Advanced Plant Habitat | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | A | 26/34 |
| Aquatic Habitat | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | A | 16/30 |
| Mice Drawer System | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | A | 23/29 |
| Vented Fly Box | ✓ | ✓ | ✓ | n/a | ✓ | ✓ | A | 16/28 |
| Advanced Biological Research System | ✓ | ✓ | ✓ | – | ✓ | ✓ | B | 18/28 |
| Avian Development Facility | ✓ | – | ✓ | – | ✓ | ✓ | B | 19/29 |
| Animal Enclosure Module | ✓ | – | ✓ | – | ✓ | ✓ | B | 21/27 |
| Animal Enclosure Module Transporter | – | – | ✓ | ✓ | ✓ | ✓ | B | 21/27 |
| BioCell Cell Culture and Plate Habitats | – | – | ✓ | n/a | ✓ | ✓ | B | 15/28 |
| Bioculture System | – | ✓ | ✓ | – | ✓ | ✓ | B | 17/32 |
| Biomass Production System | ✓ | – | ✓ | – | ✓ | ✓ | B | 18/30 |
| Biological Research in Canisters - Petri Dish | – | ✓ | ✓ | n/a | ✓ | ✓ | B | 19/33 |
| Cell Biology Experiment Facility / Plant Exper | ✓ | ✓ | ✓ | – | ✓ | ✓ | B | 16/28 |
| Cell Culture Module | – | ✓ | ✓ | n/a | ✓ | ✓ | B | 12/29 |
| Closed Equilibrated Biological Aquatic System | – | – | ✓ | ✓ | ✓ | ✓ | B | 16/30 |
| European Modular Cultivation System | ✓ | – | ✓ | – | ✓ | ✓ | B | 17/28 |
| Fungal Immunity | – | – | ✓ | n/a | ✓ | ✓ | B | 16/29 |
| Fruit Fly Lab | – | – | ✓ | n/a | ✓ | ✓ | B | 16/30 |
| Group Activation Pack - Fluid Processing Appar | – | ✓ | ✓ | n/a | ✓ | ✓ | B | 13/29 |
| JAXA Mouse Habitat Unit | – | – | ✓ | ✓ | ✓ | ✓ | B | 21/27 |
| Jellyfish Kit | – | – | ✓ | n/a | ✓ | ✓ | B | 11/29 |
| Multiple Orbital Bioreactor with Instrumentati | – | – | ✓ | n/a | ✓ | ✓ | B | 10/29 |
| Multi-use Variable-g Platform | – | ✓ | ✓ | – | ✓ | ✓ | B | 20/35 |
| Research Animal Holding Facility | ✓ | – | ✓ | ✓ | – | ✓ | B | 22/27 |
| Rodent Habitat | – | – | ✓ | ✓ | ✓ | ✓ | B | 19/27 |
| Space Automated Bioproduct Laboratory | ✓ | – | ✓ | n/a | ✓ | ✓ | B | 18/31 |
| Vegetable Production System | ✓ | – | ✓ | – | ✓ | ✓ | B | 19/29 |
| Vestibular Function Experiment Unit | – | – | ✓ | ✓ | ✓ | ✓ | B | 16/33 |
| ADvanced Space Experiment Processor | – | ✓ | – | – | ✓ | ✓ | C | 12/29 |
| BOTEX incubator | – | ✓ | – | – | ✓ | ✓ | C | 9/28 |
| Biological Research In Canisters - BRIC-100 | – | – | ✓ | n/a | – | ✓ | C | 11/27 |
| BRIC-100VC | – | – | ✓ | n/a | – | ✓ | C | 11/28 |
| BRIC-60 | – | – | ✓ | n/a | – | ✓ | C | 13/29 |
| BRIC-LED | – | ✓ | ✓ | – | – | ✓ | C | 16/30 |
| Commercial Generic Bioprocessing Apparatus | – | – | – | – | ✓ | ✓ | C | 11/29 |
| E150 Experiment Unique Equipment | – | – | – | – | – | ✓ | C | 6/27 |
| Kubik incubator | – | ✓ | – | – | ✓ | ✓ | C | 12/29 |

# Supplementary Note S4. Modelling assumptions

Table S4: Every modelling assumption in the CFD models (all other numbers reference sourced parameters).

| Model | Parameter | Reason |
|---|---|---|
| aph | obstacles[0].center | Science Carrier fills the chamber floor ('installed into the base of the growth chamber', registry 2017); footprint equal to the chamber footprint by construction (enclosure footprint was derived from the carrier). |
| aph | zones[0].size | Canopy zone spans the planted surface minus 10 mm margins; 100 mm height represents a young canopy (PH-04 plants were 14 days old at the first RH change). Sensitivity: canopy height not sourced. |
| aph | zones[0].center | Canopy zone sits directly on the Science Carrier surface (z = 51 mm). |
| aph | openings[0].size | Vent sizes not published ('Vent sizes not published', aph.yaml). 300 x 60 mm slots assumed on each ECS side. |
| aph | openings[0].center | ECS returns air 'at plant level from opposing sides' (Monje et al. 2020): inlets centred 50 mm above the carrier. |
| aph | openings[1].size | As openings[0]. |
| aph | openings[1].center | As openings[0]. |
| aph | openings[2].size | Outlet size not published; equal to inlets. |
| aph | openings[2].center | ECS fans 'remove air from the top of the growth chamber above the plants' (Monje et al. 2020): outlets centred 40 mm below the ceiling on the same side walls (ECS modules are 'mounted on each side'). |
| aph | openings[3].size | As openings[2]. |
| aph | openings[3].center | As openings[2]. |
| aph | openings.inlet_velocity_ interpretation | The PH-04 'wind speed of 0.3 m/s' set-point is applied as the inlet face velocity; where APH measures wind speed is not stated (aph.yaml note). |
| aph | zones.canopy.co2_load | Canopy net CO2 uptake uses the wheat range 7-25 umol m-2 s-1 (Wheeler 1996 as cited by Monje et al. 2020; mid-point 16) over the 0.19 m2 growth area; no pepper rate was found. Transpiration and lamp/canopy heat are not modelled (no sourced value). |
| aph_fanmax | obstacles[0].center | Science Carrier fills the chamber floor ('installed into the base of the growth chamber', registry 2017); footprint equal to the chamber footprint by construction (enclosure footprint was derived from the carrier). |
| aph_fanmax | zones[0].size | Canopy zone spans the planted surface minus 10 mm margins; 100 mm height represents a young canopy (PH-04 plants were 14 days old at the first RH change). Sensitivity: canopy height not sourced. |
| aph_fanmax | zones[0].center | Canopy zone sits directly on the Science Carrier surface (z = 51 mm). |
| aph_fanmax | openings[0].size | Vent sizes not published ('Vent sizes not published', aph.yaml). 300 x 60 mm slots assumed on each ECS side. |
| aph_fanmax | openings[0].center | ECS returns air 'at plant level from opposing sides' (Monje et al. 2020): inlets centred 50 mm above the carrier. |
| aph_fanmax | openings[1].size | As openings[0]. |
| aph_fanmax | openings[1].center | As openings[0]. |
| aph_fanmax | openings[2].size | Outlet size not published; equal to inlets. |
| aph_fanmax | openings[2].center | ECS fans 'remove air from the top of the growth chamber above the plants' (Monje et al. 2020): outlets centred 40 mm below the ceiling on the same side walls (ECS modules are 'mounted on each side'). |
| aph_fanmax | openings[3].size | As openings[2]. |
| aph_fanmax | openings[3].center | As openings[2]. |
| aph_fanmax | openings.inlet_velocity_ interpretation | The PH-04 'wind speed of 0.3 m/s' set-point is applied as the inlet face velocity; where APH measures wind speed is not stated (aph.yaml note). |

*Table S4, continued*

| Model | Parameter | Reason |
|---|---|---|
| aph_fanmax | zones.canopy.co2_load | Canopy net CO2 uptake uses the wheat range 7-25 umol m-2 s-1 (Wheeler 1996 as cited by Monje et al. 2020; mid-point 16) over the 0.19 m2 growth area; no pepper rate was found. Transpiration and lamp/canopy heat are not modelled (no sourced value). |
| aqh | openings[0].size | Water inlet/outlet geometry of the aquarium is not published (aqh.yaml: 'aquarium water inlet/outlet geometry'). 20 x 20 mm ports assumed. |
| aqh | openings[0].center | Inlet low on one end wall, outlet high on the opposite end wall (assumed cross-flow arrangement). |
| aqh | openings[1].size | As openings[0]. |
| aqh | openings[1].center | As openings[0]. |
| aqh | domain.water_only | The aquarium is modelled full of water; the air pocket of the air-water interface ('air stabilizer') and the fish are not represented (no dimensions or O2 uptake sourced). Walls held at the set-point temperature; no heat sources, so 0 g and 1 g are expected to coincide. |
| mds | zones[0].size | Mouse represented as a distributed source volume of 80 x 30 x 30 mm (adult-mouse scale) resting on the cage floor; body geometry not modelled as a solid. |
| mds | zones[0].center | Mouse at the floor centre. |
| mds | openings[0].size | Flow direction through the cage is not stated; waste filters are 'located below each Habitat' (Liu 2009, mds.yaml note), so air is taken to enter through the whole top grid and leave through the whole floor grid. |
| mds | openings[0].center | As openings[0].size. |
| mds | openings[1].size | As openings[0].size. |
| mds | openings[1].center | As openings[0].size. |
| mds | openings.velocity_interpretation | Cancedda 2012: airflow of 0.2-0.3 m/s through the cages; applied as a uniform face velocity (mid-point 0.25 m/s). |
| mds | environment.co2_reference | In-cage CO2 is controlled but not reported; supply air is set to the ISS U.S. Lab cabin CO2 measured during RR-1 (Choi et al. 2020; mid-point of 788-4690 ppm). MDS flew in 2009 in JEM, so this is a proxy. |
| mds | zones.mouse.co2_load | Organism-level C57BL/6 rates (ICES-2014-101) and BALB/c heat and moisture (Riskowski 2006) scaled by the generic 30 g adult mass from Cancedda 2012; MDS flew C57BL/10J males. |
| vented_fly_box_vent20 | obstacles[0].center | 15 vials (24 x 75 mm) fit the 94 x 94 x 130 mm box only lying along x in a 3 x 5 stack (upright, at most 12 fit); g-load at launch was directed into the food at the vial bottoms (OSD-347). Vials treated as solid (food + flies inside not resolved). |
| vented_fly_box_vent20 | obstacles[0].repeat.n | 3 x 5 stack, see centre. |
| vented_fly_box_vent20 | obstacles[0].repeat.pitch | Vials spaced 32 mm (y) and 25 mm (z) to fill the box cross-section. |
| vented_fly_box_vent20 | zones[0].H | CO2 leaves each vial through its open foam plug (OSD-347; Researcher's Guide); released in a 4 mm layer at the plug end (x = 84.5-88.5 mm). |
| vented_fly_box_vent20 | zones[0].center | Plug end of each vial. |
| vented_fly_box_vent20 | zones[0].repeat.n | As obstacles. |
| vented_fly_box_vent20 | zones[0].repeat.pitch | As obstacles. |
| vented_fly_box_vent20 | zones.plug.sources.CO2.count | 13 flies per vial (OSD-347: '8 virgin females and 5 male adult virgin flies') x 15 vials = 195 adults; progeny that emerge during the mission (20-60 per vial on return) are not included, so the load is a lower bound. |
| vented_fly_box_vent20 | openings[0].size | Vent hole number, size and position are not published (vented_fly_box.yaml). Run at two sizes, 20 x 20 and 60 x 60 mm per end wall, to show the sensitivity. |
| vented_fly_box_vent20 | openings[0].center | Vents centred on the end walls. |

*Table S4, continued*

| Model | Parameter | Reason |
|---|---|---|
| vented_fly_box_vent20 | openings[1].size | As openings[0]. |
| vented_fly_box_vent20 | openings[1].center | As openings[0]. |
| vented_fly_box_vent20 | openings.exterior | The exterior at each vent is taken as well mixed ISS cabin air (CO2 mid-point of the RR-1 U.S. Lab range, Choi et al. 2020) at the box temperature. |
| vented_fly_box_vent60 | obstacles[0].center | 15 vials (24 x 75 mm) fit the 94 x 94 x 130 mm box only lying along x in a 3 x 5 stack (upright, at most 12 fit); g-load at launch was directed into the food at the vial bottoms (OSD-347). Vials treated as solid (food + flies inside not resolved). |
| vented_fly_box_vent60 | obstacles[0].repeat.n | 3 x 5 stack, see centre. |
| vented_fly_box_vent60 | obstacles[0].repeat.pitch | Vials spaced 32 mm (y) and 25 mm (z) to fill the box cross-section. |
| vented_fly_box_vent60 | zones[0].H | CO2 leaves each vial through its open foam plug (OSD-347; Researcher's Guide); released in a 4 mm layer at the plug end (x = 84.5-88.5 mm). |
| vented_fly_box_vent60 | zones[0].center | Plug end of each vial. |
| vented_fly_box_vent60 | zones[0].repeat.n | As obstacles. |
| vented_fly_box_vent60 | zones[0].repeat.pitch | As obstacles. |
| vented_fly_box_vent60 | zones.plug.sources.CO2.count | 13 flies per vial (OSD-347: '8 virgin females and 5 male adult virgin flies') x 15 vials = 195 adults; progeny that emerge during the mission (20-60 per vial on return) are not included, so the load is a lower bound. |
| vented_fly_box_vent60 | openings[0].size | Vent hole number, size and position are not published (vented_fly_box.yaml). Run at two sizes, 20 x 20 and 60 x 60 mm per end wall, to show the sensitivity. |
| vented_fly_box_vent60 | openings[0].center | Vents centred on the end walls. |
| vented_fly_box_vent60 | openings[1].size | As openings[0]. |
| vented_fly_box_vent60 | openings[1].center | As openings[0]. |
| vented_fly_box_vent60 | openings.exterior | The exterior at each vent is taken as well mixed ISS cabin air (CO2 mid-point of the RR-1 U.S. Lab range, Choi et al. 2020) at the box temperature. |

# Supplementary Note S5. Discretisation uncertainty

Table S5: Grid convergence [1] of specimen-zone and domain quantities (excess over the supply state; age of air in s; $U_{max}$ in $\mathrm{m\,s^{-1}}$).

| Model | $g$ | Quantity | Fine | Medium | Coarse | $p$ | GCI (%) | Temporal sd | Convergence |
|---|---|---|---|---|---|---|---|---|---|
| MDS | 0g | zone mouse CO2 | 14.76 | 15.58 | 17.34 | 2.29 | 5.83 | 0.00 | monotonic |
| MDS | 0g | zone mouse T | 0.28 | 0.29 | 0.33 | 2.27 | 6.00 | 0.00 | monotonic |
| MDS | 0g | zone mouse AoA | 0.27 | 0.27 | 0.27 | 1.75 | 0.53 | 0.00 | monotonic |
| MDS | 0g | domain CO2 | 1.19 | 1.21 | 1.24 | 1.56 | 2.75 | 0.00 | monotonic |
| MDS | 0g | domain AoA | 0.18 | 0.18 | 0.18 | 3.12 | 0.39 | $2.79 \times 10^{-17}$ | monotonic |
| MDS | 0g | U_max_late_m_s | 0.31 | 0.30 | 0.30 | 4.44 | 0.32 | – | oscillatory |
| MDS | 1g | zone mouse CO2 | 14.79 | 15.61 | 17.38 | 2.29 | 5.86 | 0.00 | monotonic |
| MDS | 1g | zone mouse T | 0.28 | 0.29 | 0.33 | 2.26 | 6.03 | 0.00 | monotonic |
| MDS | 1g | zone mouse AoA | 0.27 | 0.27 | 0.27 | 1.81 | 0.51 | $4.92 \times 10^{-10}$ | monotonic |
| MDS | 1g | domain CO2 | 1.19 | 1.21 | 1.24 | 1.57 | 2.76 | 0.00 | monotonic |
| MDS | 1g | domain AoA | 0.18 | 0.18 | 0.18 | 3.11 | 0.40 | 0.00 | monotonic |
| MDS | 1g | U_max_late_m_s | 0.31 | 0.30 | 0.30 | 4.27 | 0.34 | – | oscillatory |
| APH | 0g | zone canopy CO2 | -6.43 | -6.50 | -6.50 | 10.35 | 0.04 | 0.24 | monotonic (within unsteadiness) |
| APH | 0g | zone canopy AoA | 1.97 | 1.98 | 1.97 | 3.24 | 0.10 | 0.11 | oscillatory (within unsteadiness) |
| APH | 0g | domain CO2 | -7.19 | -7.14 | -7.22 | 1.40 | 1.42 | 0.08 | oscillatory (within unsteadiness) |
| APH | 0g | domain AoA | 4.91 | 4.89 | 4.94 | 4.31 | 0.08 | 0.09 | oscillatory (within unsteadiness) |
| APH | 0g | U_max_late_m_s | 0.45 | 0.45 | 0.43 | 2.95 | 0.79 | – | monotonic |
| APH | 1g | zone canopy CO2 | -6.64 | -6.82 | -6.46 | 1.96 | 3.71 | 0.27 | oscillatory (within unsteadiness) |
| APH | 1g | zone canopy AoA | 2.10 | 2.23 | 2.04 | 1.24 | 14.36 | 0.17 | oscillatory (within unsteadiness) |
| APH | 1g | domain CO2 | -7.22 | -7.31 | -7.23 | 0.50 | 8.51 | 0.06 | oscillatory (within unsteadiness) |
| APH | 1g | domain AoA | 5.00 | 5.05 | 4.95 | 1.94 | 1.39 | 0.08 | oscillatory (within unsteadiness) |
| APH | 1g | U_max_late_m_s | 0.45 | 0.44 | 0.44 | 0.52 | 11.69 | – | monotonic |
| VFB (20 mm) | 0g | zone plug CO2 | 184.46 | 187.94 | 197.02 | 2.87 | 2.27 | $6.56 \times 10^{-3}$ | monotonic |
| VFB (20 mm) | 0g | zone plug AoA | 460.83 | 469.19 | 496.32 | 3.65 | 1.54 | 0.02 | monotonic |
| VFB (20 mm) | 0g | domain CO2 | 145.41 | 148.53 | 157.31 | 3.14 | 2.27 | $7.00 \times 10^{-3}$ | monotonic |
| VFB (20 mm) | 0g | domain AoA | 490.81 | 499.40 | 527.38 | 3.67 | 1.47 | 0.02 | monotonic |

# Supplementary Note S6. Geometry of the modelled hardware

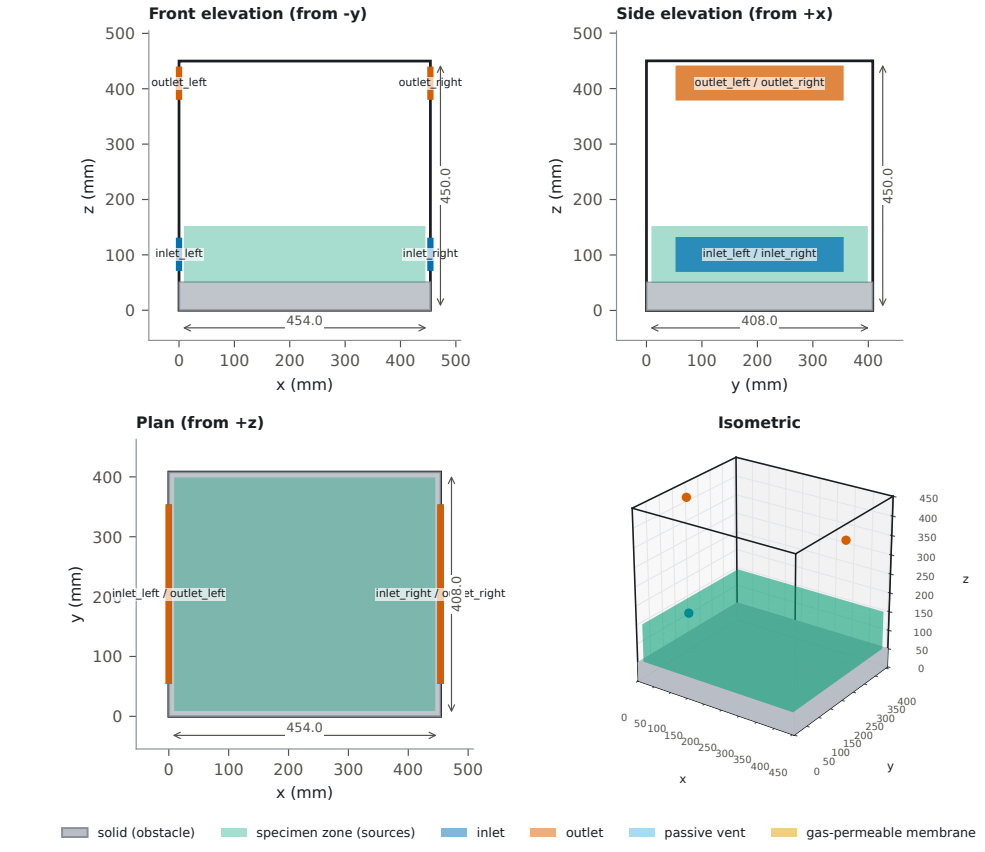



Figure S1: Advanced Plant Habitat growth chamber as modelled (PH-04 set-points).

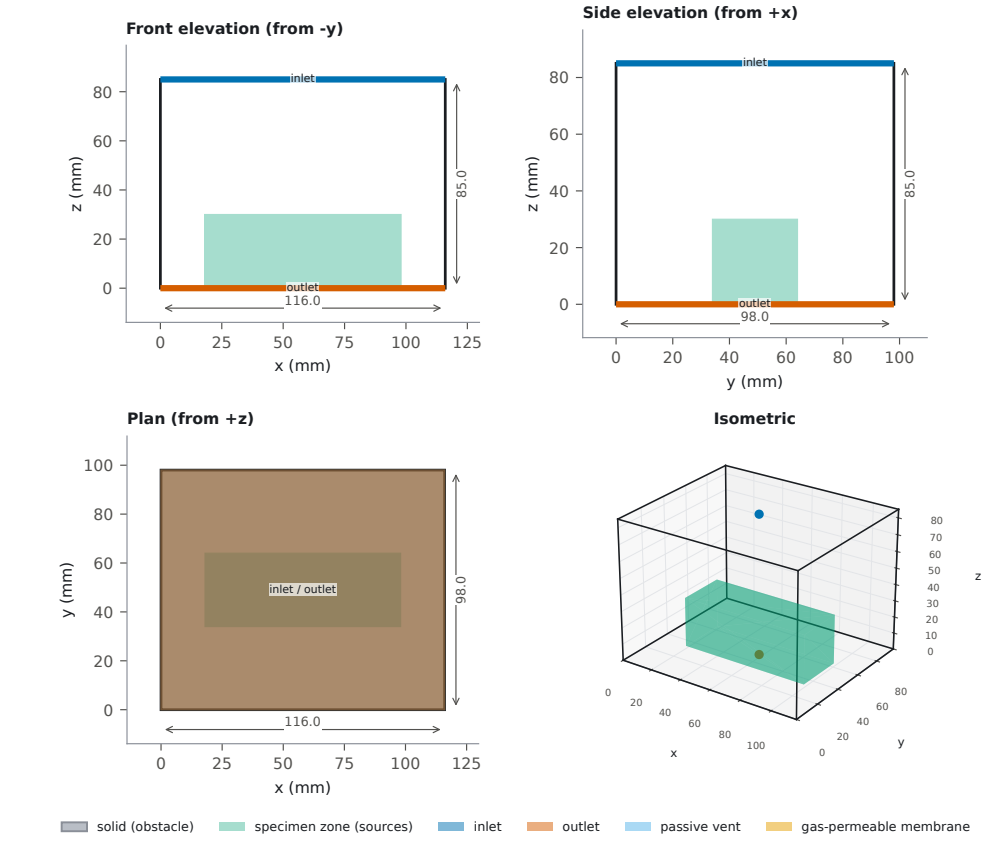



Figure S2: One Mice Drawer System cage as modelled.

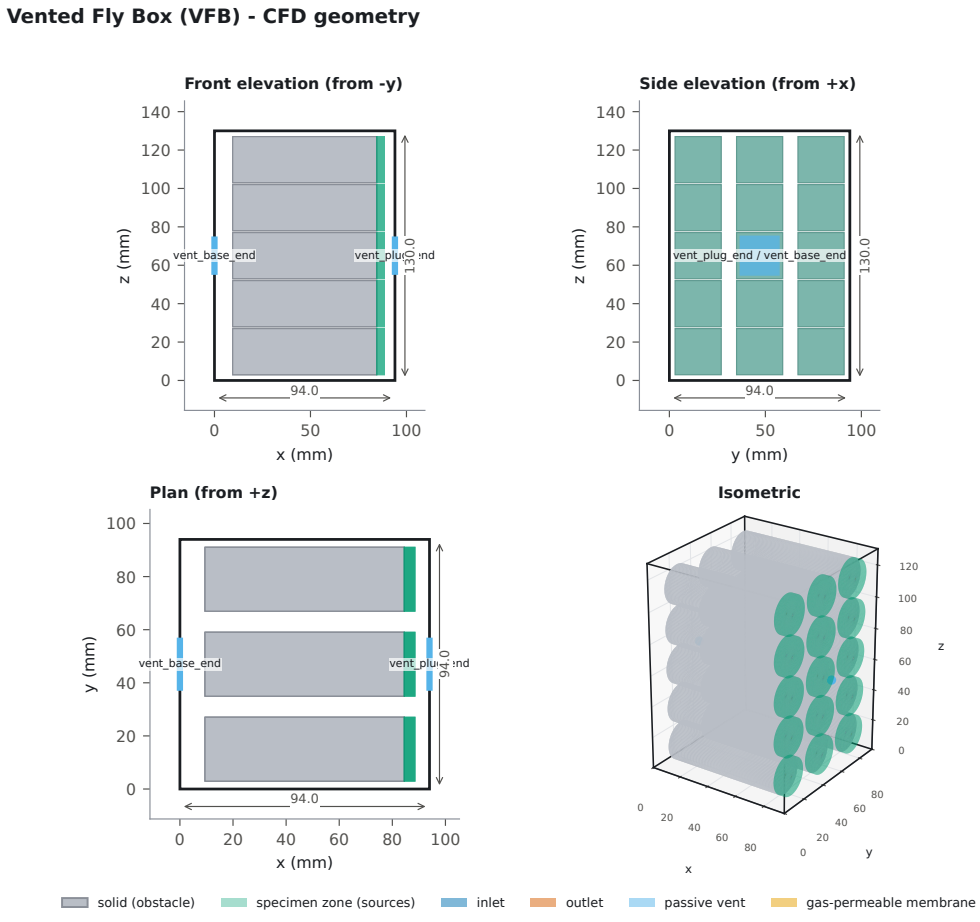



Figure S3: Vented Fly Box as modelled (20 mm vents; the 60 mm variant differs only in vent size).

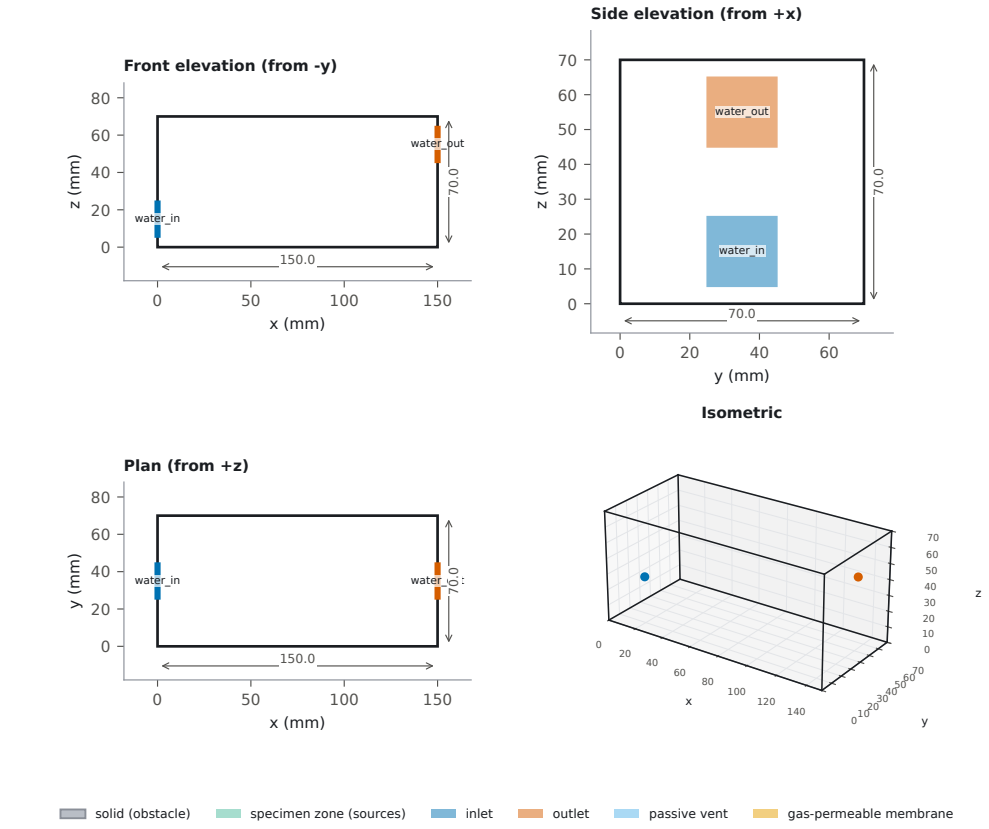



Figure S4: One Aquatic Habitat aquarium as modelled (water).

## Supplementary Note S7. Field slices

mds: vertical mid-plane, flight (0 g) vs ground control (1 g)

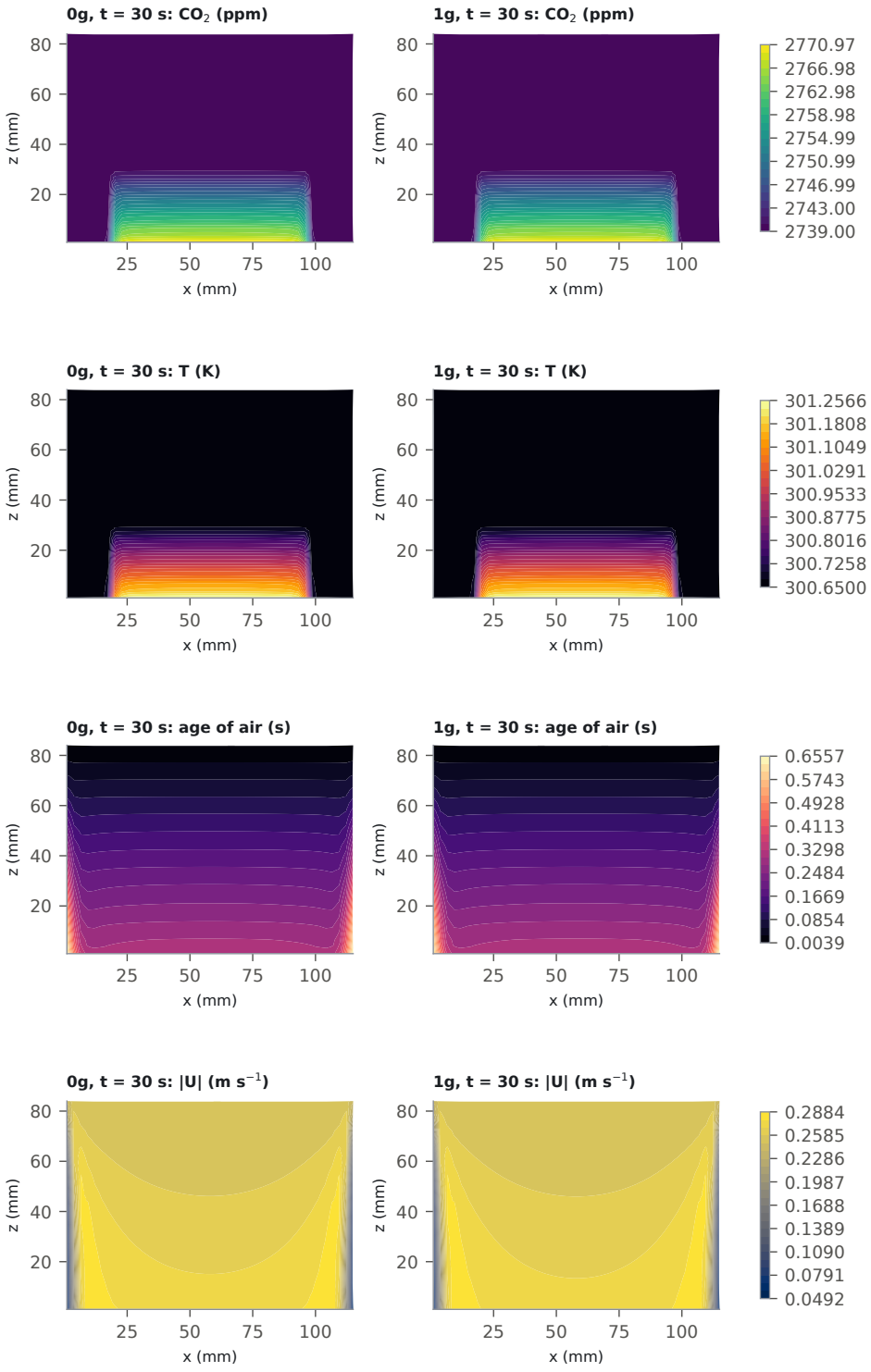



Figure S5: MDS cage, vertical mid-plane at the end of the run, $0\,g$ and $1\,g$.

aph: vertical mid-plane, flight (0 g) vs ground control (1 g)

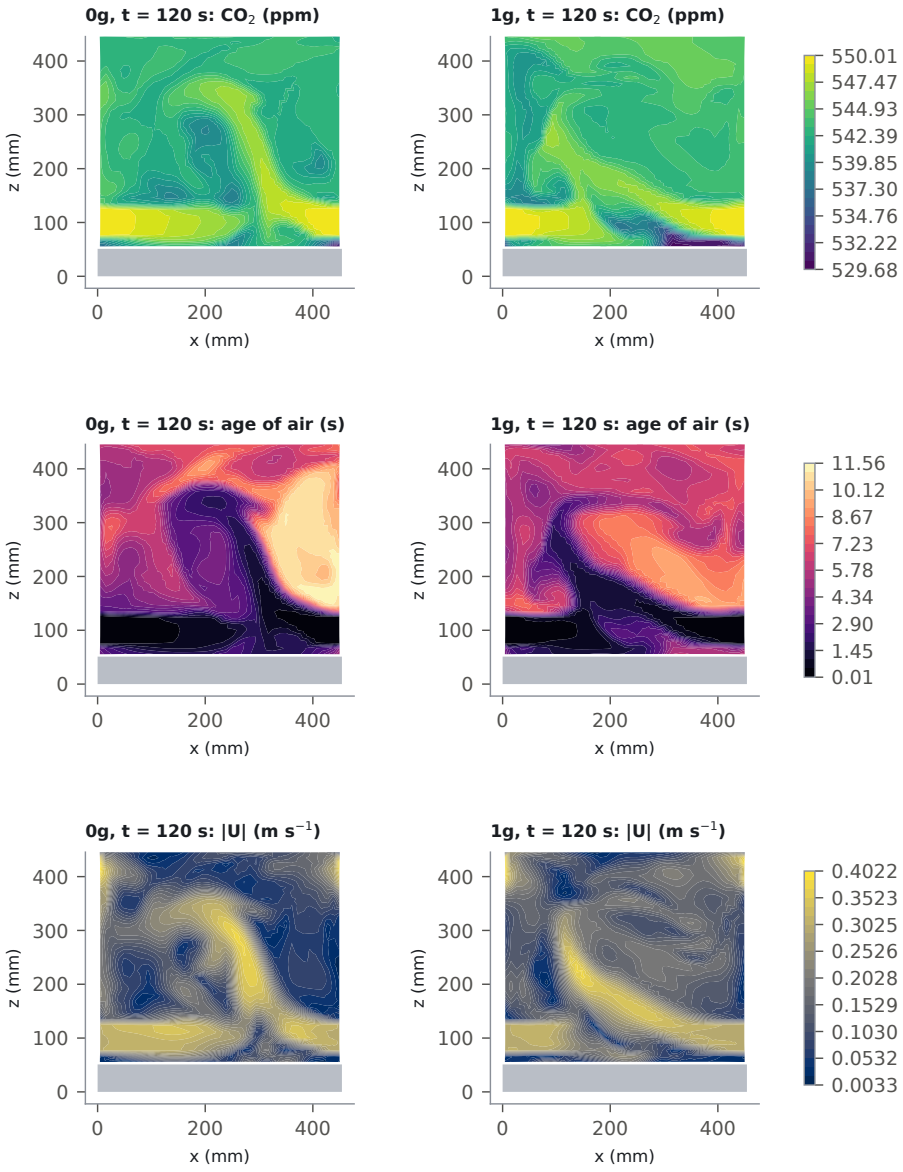



Figure S6: APH growth chamber, vertical mid-plane at the end of the run, $0\,g$ and $1\,g$.

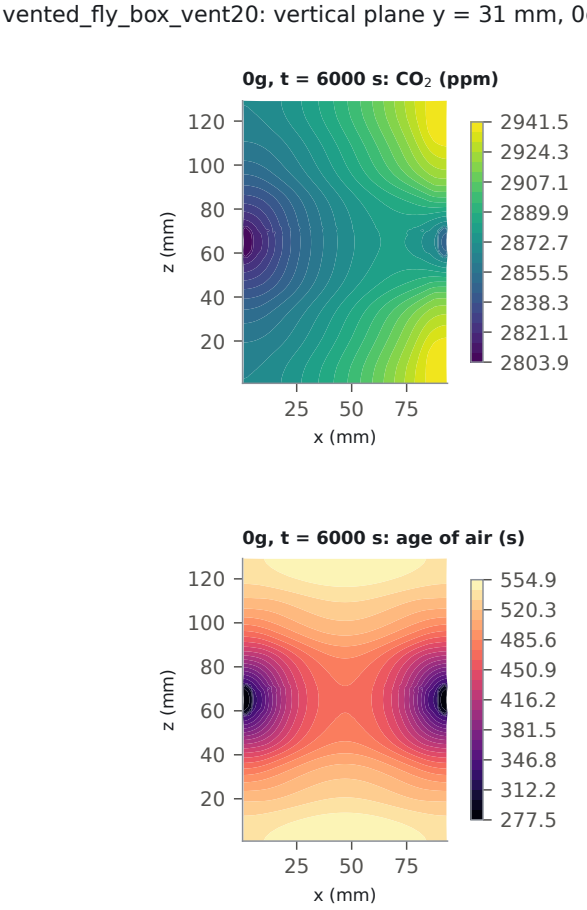



Figure S7: Vented Fly Box (20 mm vents), vertical mid-plane at the end of the run, 0 $g$ and 1 $g$.

aqh: vertical mid-plane, 0g only (1 g identical by construction)

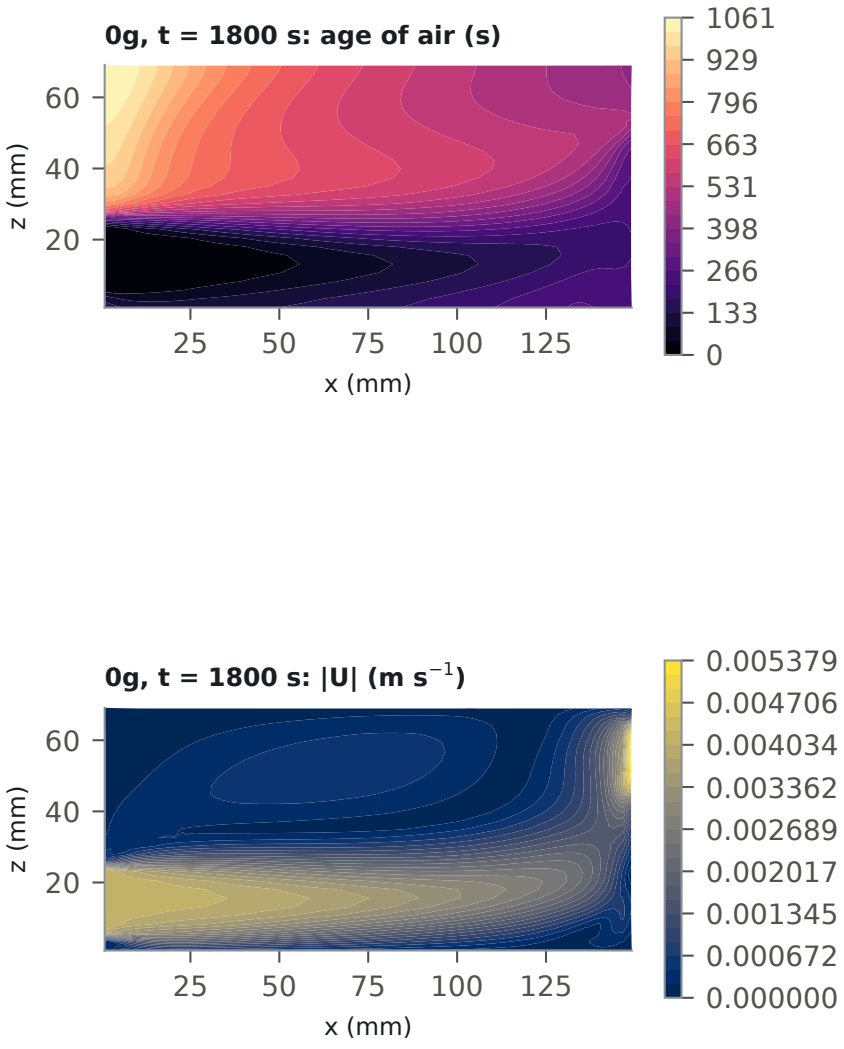



Figure S8: AQH aquarium, vertical mid-plane at the end of the run, $0\,g$ and $1\,g$.

# Supplementary Note S8. Environment cards

Time series of specimen-zone excess over the supply or initial state, $0\,g$ (blue) and $1\,g$ (red), medium meshes.

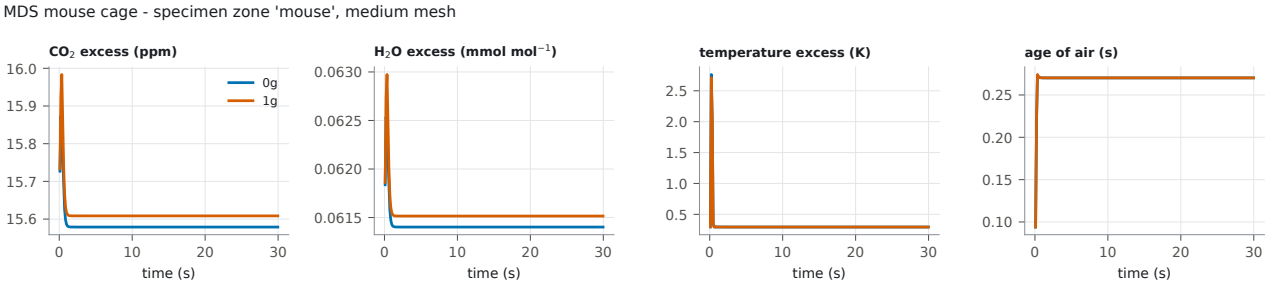



Figure S9: MDS cage.

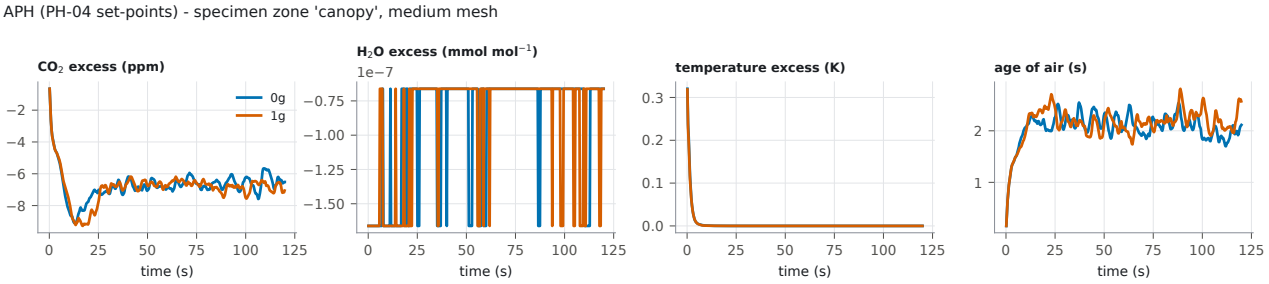



Figure S10: APH, PH-04 set-points.

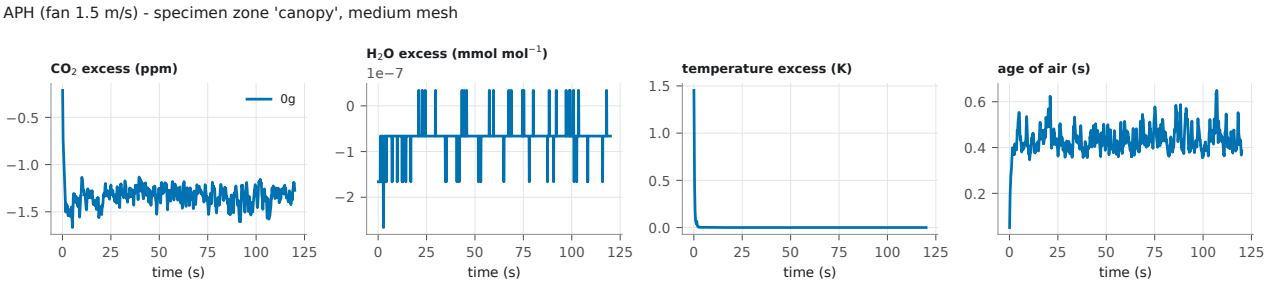



Figure S11: APH, fan at 1.5 m/s.

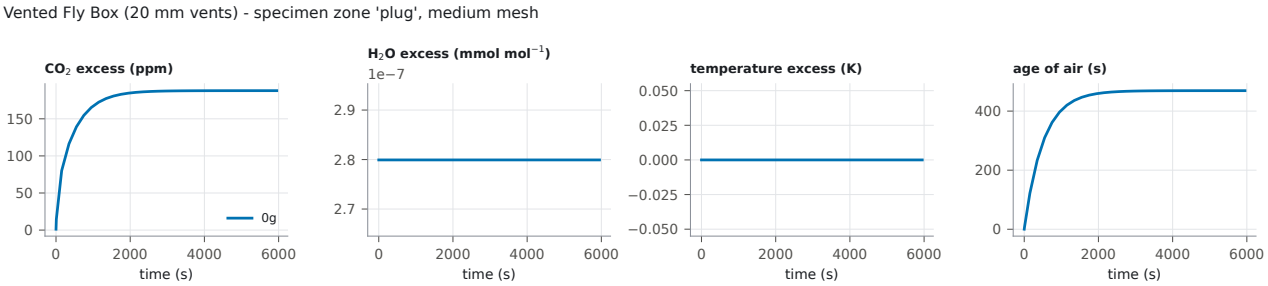



Figure S12: Vented Fly Box, 20 mm vents.

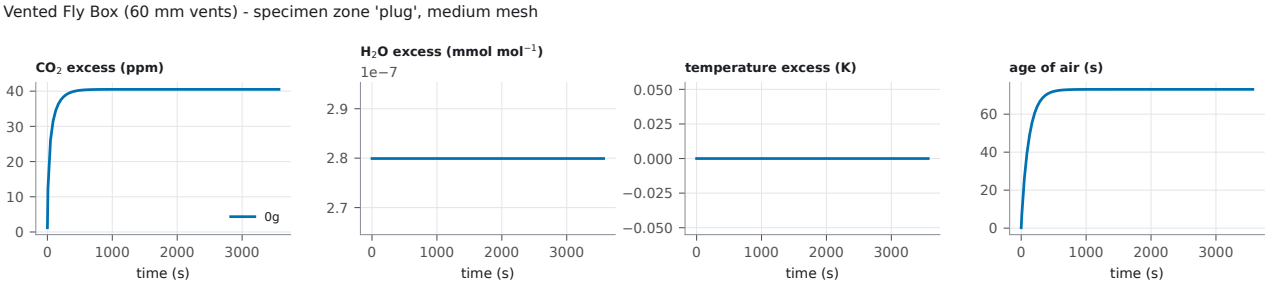



Figure S13: Vented Fly Box, 60 mm vents.

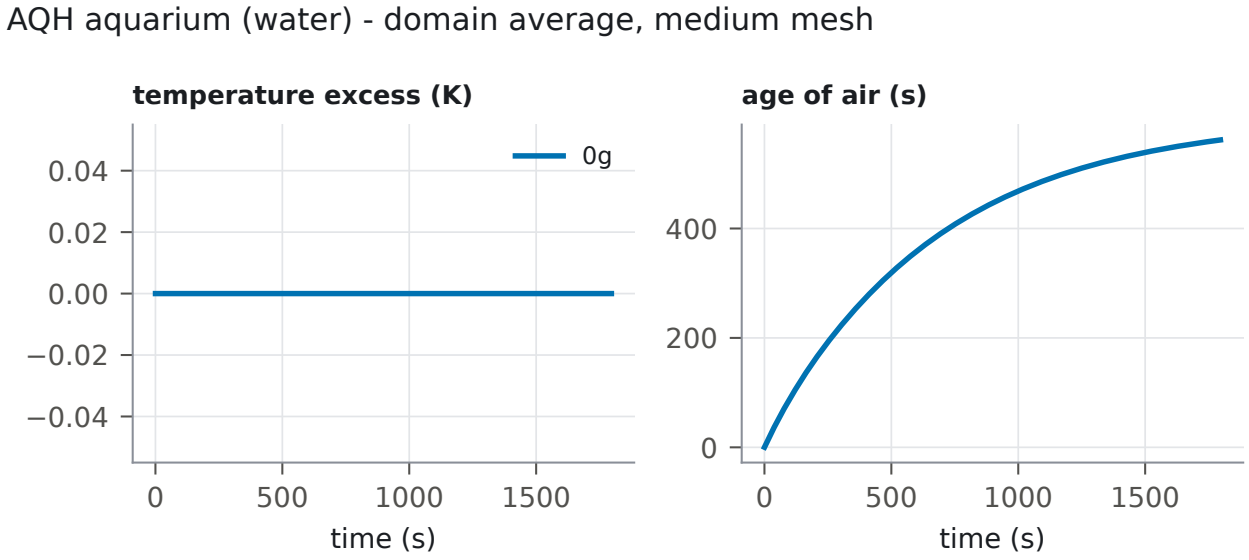



Figure S14: AQH aquarium (water).

## References

[1] Ismail B. Celik, Urmila Ghia, Patrick J. Roache, and Christopher J. Freitas. Procedure for estimation and reporting of uncertainty due to discretization in CFD applications. *Journal of Fluids Engineering*, 130:078001, 2008. doi: 10.1115/1.2960953.